📊 QUANT FINANCE INTERVIEW QUESTIONS

🔹 SECTION 1: Probability & Statistics (Q1–Q12)

Q1. What is a random variable?

Answer:
A random variable assigns a numerical value to each outcome of a random experiment.

Explanation:

- Example: Tossing a coin
  - Head → 1
  - Tail → 0

Used to model uncertainty mathematically.

Q2. Difference between discrete and continuous random variables?

Discrete	Continuous
Finite/countable	Infinite values
PMF	PDF
Dice roll	Stock price

Q3. What is expectation (mean)?

Answer:
Expected value is the weighted average of all possible outcomes.

$$E[X] = \sum x_i p_i$$

Explanation:
In finance, expectation represents average return.

Q4. What is variance? Why is it important?

Answer:
Variance measures how much values deviate from the mean.

$$\boldsymbol{Var(X) = E[(X - \mu)^2]}$$

Explanation:
In finance, variance = risk.

Q5. What is standard deviation?

Answer:
Square root of variance.

Explanation:

- Same units as original data
- Easier to interpret risk

Q6. What is covariance?

Answer:
Measures how two variables move together.

- Positive → move together
- Negative → move opposite

Q7. What is correlation?

Answer:
Standardized covariance between −1 and +1.

Explanation:
Used to measure diversification benefits.

Q8. What is a normal distribution?

Answer:
A symmetric bell-shaped distribution.

Key Properties:

- Mean = Median = Mode
- 68–95–99.7 rule

Q9. Why is normal distribution important in finance?

Explanation:

- Used in option pricing
- Used in risk models
- Simplifies mathematical modeling

⚠️ But real markets have fat tails

Q10. What is the Law of Large Numbers?

Answer:
As sample size increases, sample mean → true mean.

Finance intuition:
Long-term averages stabilize.

Q11. What is Central Limit Theorem (CLT)?

Answer:
Sum of random variables tends toward normal distribution.

Why important:
Justifies using normal distribution in finance.

Q12. What is skewness and kurtosis?

- Skewness → asymmetry
- Kurtosis → tail risk

Finance data often:

- Negative skew
- High kurtosis

🔹 SECTION 2: Financial Basics (Q13–Q22)

Q13. What is return?

$$\boldsymbol{Return = \frac{P_t - P_{t-1}}{P_{t-1}}}$$

Measures gain or loss over time.

Q14. Difference between simple return and log return?

Simple	Log
Easy	Time-additive
Less stable	Used in quant models

Q15. What is risk-free rate?

Answer:
Return with zero default risk.

Example:
Government treasury bills.

Q16. What is compounding?

$$\boldsymbol{FV = PV(1 + r)^n}$$

Interest earns interest.

Q17. What is present value (PV)?

Discount future cash flows to today.

Q18. What is Net Present Value (NPV)?

Answer:
Sum of discounted cash flows minus initial investment.

Used for investment decisions.

Q19. What is arbitrage?

Answer:
Risk-free profit with zero net investment.

Key principle:
Markets eliminate arbitrage opportunities quickly.

Q20. What is no-arbitrage principle?

Answer:
Two identical assets must have the same price.

Foundation of pricing models.

Q21. What is volatility?

Answer:
Standard deviation of returns.

Represents uncertainty.

Q22. Historical vs implied volatility?

Historical	Implied
Past data	Market expectation
Realized	Forward-looking

- SECTION 3: Portfolio Theory & Risk (Q23–Q32)

Q23. What is diversification?

Answer:
Reducing risk by combining uncorrelated assets.

Q24. What is efficient frontier?

Answer:
Set of portfolios with:

- Maximum return for given risk
- Minimum risk for given return

Q25. What is Markowitz portfolio theory?

Uses:

- Expected return
- Variance
- Covariance

to optimize portfolios.

Q26. What is systematic vs unsystematic risk?

Systematic	Unsystematic
Market risk	Asset-specific
Cannot diversify	Can diversify

Q27. What is Beta?

$$\boldsymbol{\beta} = \frac{\boldsymbol{Cov(asset, market)}}{\boldsymbol{Var(market)}}$$

Measures sensitivity to market.

Q28. What is CAPM?

$$\boldsymbol{E(R) = R_f + \beta(R_m - R_f)}$$

Links risk and expected return.

Q29. What is Value at Risk (VaR)?

Answer:
Maximum expected loss at given confidence level.

Example:

1-day 95% VaR = $1M

Q30. Difference between VaR and CVaR?

VaR	CVaR
Threshold	Average loss beyond VaR
Ignores tail	Captures tail risk

Q31. What is drawdown?

Peak-to-trough loss.

Q32. What is Sharpe Ratio?

$$Sharpe = \frac{R_p - R_f}{\sigma}$$

Measures risk-adjusted return.

- SECTION 4: Time Series & Markets (Q33–Q40)

Q33. What is a time series?

Sequence of data points over time.

Q34. Stationary vs non-stationary series?

Stationary	Non-stationary
Constant mean & variance	Changing over time

Q35. Why stationarity matters?

Most models assume stationarity.

Q36. What is autocorrelation?

Correlation of a series with its past values.

Q37. What is white noise?

Random series with:

- Zero mean
- No correlation

Q38. What is mean reversion?

Prices tend to return to average.

Q39. What is random walk?

Price changes are unpredictable.

Q40. Efficient Market Hypothesis (EMH)?

Prices reflect all available information.

- SECTION 5: Derivatives & Trading Intuition (Q41–Q50)

Q41. What is a derivative?

Value depends on underlying asset.

Q42. Call vs Put option?

Call	Put
Buy	Sell
Bullish	Bearish

Q43. What are option Greeks?

- Delta → price sensitivity
- Gamma → delta change
- Vega → volatility
- Theta → time decay
- Rho → interest rate

Q44. What is Black–Scholes model?

Option pricing model assuming:

- Lognormal prices
- Constant volatility
- No arbitrage

Q45. What is implied volatility smile?

Volatility differs across strikes.

Shows market fear.

Q46. What is bid-ask spread?

Difference between buy and sell price.

Q47. What is liquidity?

Ease of trading without price impact.

Q48. What is slippage?

Difference between expected and executed price.

Q49. What is backtesting?

Testing strategy on historical data.

Q50. What is overfitting in trading strategies?

Model fits noise, not signal.

Dangerous in live markets.